

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JnanaSangama”, Belgaum -590014, Karnataka.



LAB REPORT
on

OPERATING SYSTEMS

(23CS4PCOPS)

Submitted by

SHASHANK SHANTHARAM NAYAK (1BM23CS313)

in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING

in
COMPUTER SCIENCE AND ENGINEERING



B.M.S. COLLEGE OF ENGINEERING

(Autonomous Institution under VTU)

BENGALURU-560019

Feb-2025 to June-2025

B. M. S. College of Engineering,
Bull Temple Road, Bangalore 560019 (Affiliated To Visvesvaraya
Technological University, Belgaum) **Department of Computer
Science and Engineering**



CERTIFICATE

This is to certify that the Lab work entitled “OPERATING SYSTEMS – 23CS4PCOPS” carried out by SHASHANK SHANTHARAM NAYAK (1BM23CS313), who is bonafide student of **B. M. S. College of Engineering**. It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2024. The Lab report has been approved as it satisfies the academic requirements in respect of a **OPERATING SYSTEMS - (23CS4PCOPS)** work prescribed for the said degree.

Amruta B.

Assistant Professor
Department of CSE,
BMSCE, Bengaluru

Dr. Kavitha Sooda

Professor and Head Department of CSE

BMSCE, Bengaluru

Index Sheet

Sl. No.	Experiment Title	Page No.
1.	Write a C program to simulate the following CPU scheduling algorithm to find turnaround time and waiting time. (Any one) a) FCFS b) SJF	
2.	Write a C program to simulate multi-level queue scheduling algorithm considering the following scenario. All the processes in the system are divided into two categories –system processes and user processes. System processes are to be given higher priority than user processes. Use FCFS scheduling for the processes in each queue.	
3.	Write a C program to simulate Real-Time CPU Scheduling algorithms a) Rate- Monotonic	
4.	Write a C program to simulate: a) Producer-Consumer problem using semaphores. b) Dining-Philosopher's problem	
5.	Write a C program to simulate: a) Bankers' algorithm for the purpose of deadlock avoidance.	
6.	Write a C program to simulate the following contiguous memory allocation techniques. a) Worst-fit b) Best-fit c) First-fit	
7.	Write a C program to simulate page replacement algorithms. a) FIFO b) LRU c) Optimal	
8.	Write a C program to simulate the following file allocation strategies. a) Sequential b) Indexed c) Linked	

Course Outcome

CO1	Apply the different concepts and functionalities of Operating System
CO2	Analyze various Operating system strategies and techniques
CO3	Demonstrate the different functionalities of Operating System
CO4	Conduct practical experiments to implement the functionalities of Operating system